- ***Response time*** —If your response time isn't consistent, then your application isn't reliable.
- ***Error rate*** —By monitoring your logs, you can see how many errors you have over time.

You could have an application that is up all the time (highly available) but doesn't function correctly 50% of the time (low reliability). You could also have an application that does what it's supposed to do all the time (highly reliable) but is down for maintenance for an hour a day (low availability).

Even though the two are different, there's overlap in how they're measured. Two products that help measure uptime are Hyperspin (www.hyperspin.com/en/) and New Relic (www.newrelic.com/). Example reports from Hyperspin in figure 8.14 show the data you need to measure availability.

Figure 8.14. Example basic report from Hyperspin

Period	Uptime	(incl. maintenance)	Downtime	Outage	Outage History	Detailed Log
2015 Yearly Total	99.751%	99.751%	01hr 25min	17	Outage History	Detailed Log
2015 Jan	99.751%	99.751%	01hr 25min	17	Outage History	Detailed Log
2014 Yearly Total	99.787%	99.787%	18hr 40min	61	Outage History	Detailed Log
2014 Dec	97.782%	97.782%	16hr 30min	42	Outage History	Detailed Log
2014 Nov	99.907%	99.907%	40min	6	Outage History	-
2014 Oct	99.877%	99.877%	55min	10	Outage History	-
2014 Sep	99.954%	99.954%	20min	2	Outage History	-
2014 Aug	99.966%	99.966%	15min	1	Outage History	-
2014 Jul	100.000%	100.000%	00min	0	Outage History	-

As you can see, you have uptime, downtime, and the number of outages, and you can click into the report to get details on outage history.

New Relic has a similar feature that creates an availability report. The New Relic version is shown in figure 8.15.

Figure 8.15. New Relic's version of visualizing and reporting availability